

Credit Risk Modeling

Predicting Mortgage Defaults in MBS Pools

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The Challenge



Market Scale

\$12 Trillion

MBS Market Size



Crisis Impact

2008 Financial Crisis

Triggered by underestimated
defaults



**Goal: Predict loan-level defaults with
accuracy and transparency**

Approach: Logistic Regression



Regulatory Acceptance

Interpretable for compliance



Coefficient Explainability

Clear risk drivers



Production Ready

Easy calibration & deployment

Data: Freddie Mac Loan Performance



20.32M

Records



20

Features Engineered



0.32%

Default Rate



Period: 2014 - March 2025 | ⚠ Challenge: Extreme Class Imbalance

Rigorous Validation Strategy

Training

≤ Dec 2023

7.87M records

Validation

Jan-Jun 2024

295K records

Test

> Jul 2024

12.16M records

✓ **Key Strength:** True out-of-time testing on completely unseen future data

Feature Engineering

Borrower Profile

- Credit Score
- Original LTV
- DTI Ratio
- Number of Borrowers

Loan Characteristics

- Interest Rate
- Loan Age
- Remaining Term Ratio

Risk Flags

- High DTI (>45%)
- High LTV (>95%)
- Low FICO (<660)
- Negative Equity

Macro at Origination

- 30-yr Mortgage Rate
- Unemployment
- HPI YoY Change
- Fed Funds Rate

Modeling Pipeline

1

Preprocessing

Median imputation → Standard scaling → One-hot encoding



2

Model Training

Logistic Regression with L2 regularization (class_weight=None, solver='lbfgs')



3

Hyperparameter Tuning

Cross-validated grid search on training data



4

Calibration

Platt scaling on validation set for reliable probabilities

Out-of-Time Performance (Jul 2024+)

Test: 12.16M loans | Default Rate: 0.32%

82.8%

ROC-AUC

2.01%

Precision
@ 95th %ile

31.0%

Recall
@ 95th %ile

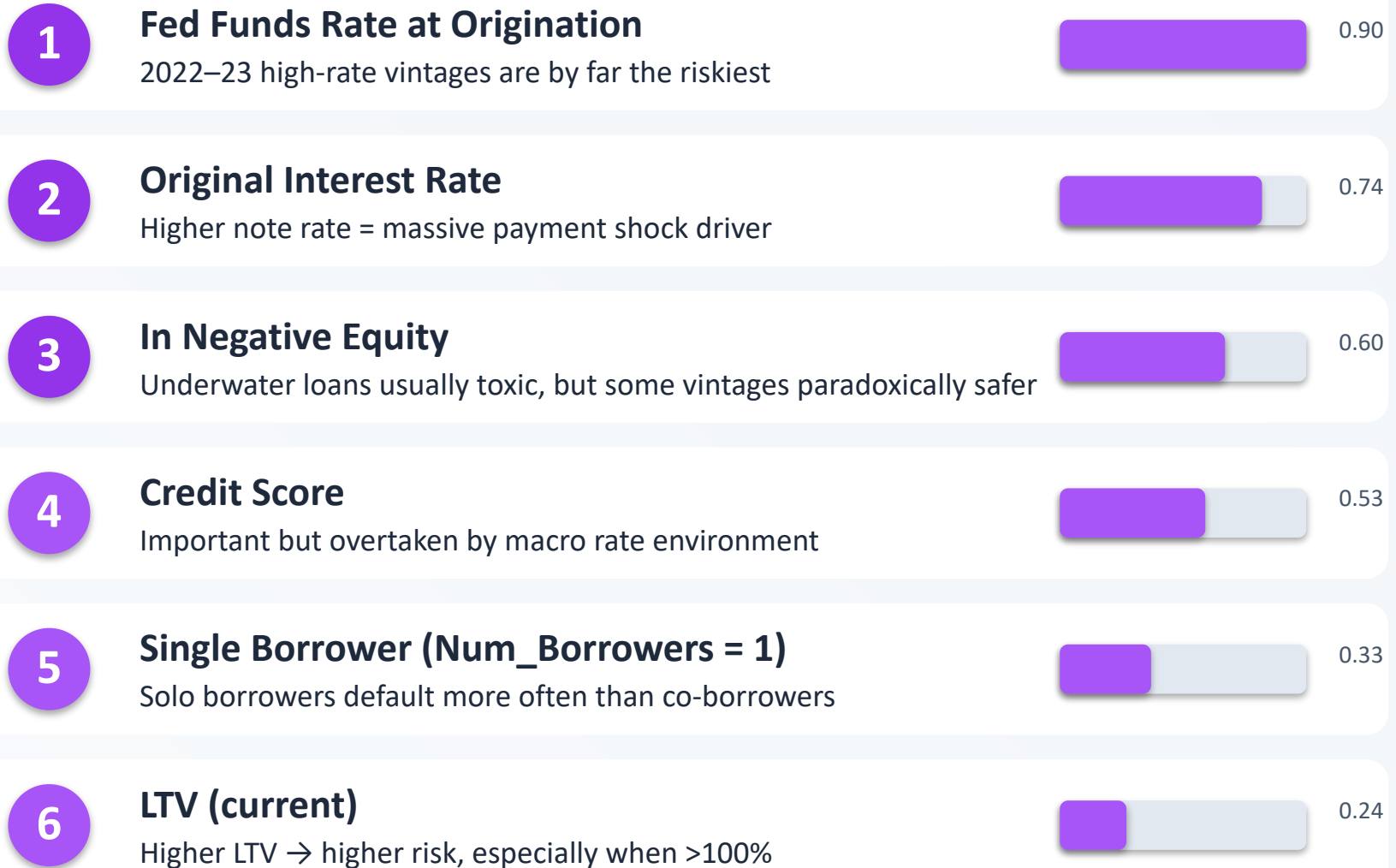
6.2×

Lift vs
Baseline



Extremely strong ranking power sustained into 2025

Key Risk Drivers (SHAP Analysis)



Macro rate environment now dominates over pure credit score

0.23

Economic Insights



2021-2023 vintages (high-rate environment) are by far the riskiest



Macro conditions at origination matter more than borrower credit score



Being in negative equity is the 3rd strongest driver of default risk



Judicial foreclosure states show moderately higher risk



Credit score drops to 4th strongest driver (from 6th in initial model)

Limitations & Future Enhancements



Current Limitations

- Current macro variables not incorporated (only origination)
- No competing risk modeling with prepayment
- Missing delinquency burnout and payment shock features

✓ Future Enhancements

- Add dynamic unemployment and HPI changes
- Implement LightGBM/XGBoost for higher AUC
- Multi-period survival or cure modeling
- Incorporate prepayment as competing risk

Key Takeaways

Built transparent default model on 20M+ real loan-month records

True out-of-time ROC-AUC 82.8% on 2024 H2-2025 Q1 data

Top-5% precision 2.01% (6.2× lift) at only 0.32% baseline default rate

Macro origination conditions (Fed Funds & mortgage rates) dominate risk

Production-ready: pickled pipeline + 95th percentile threshold saved

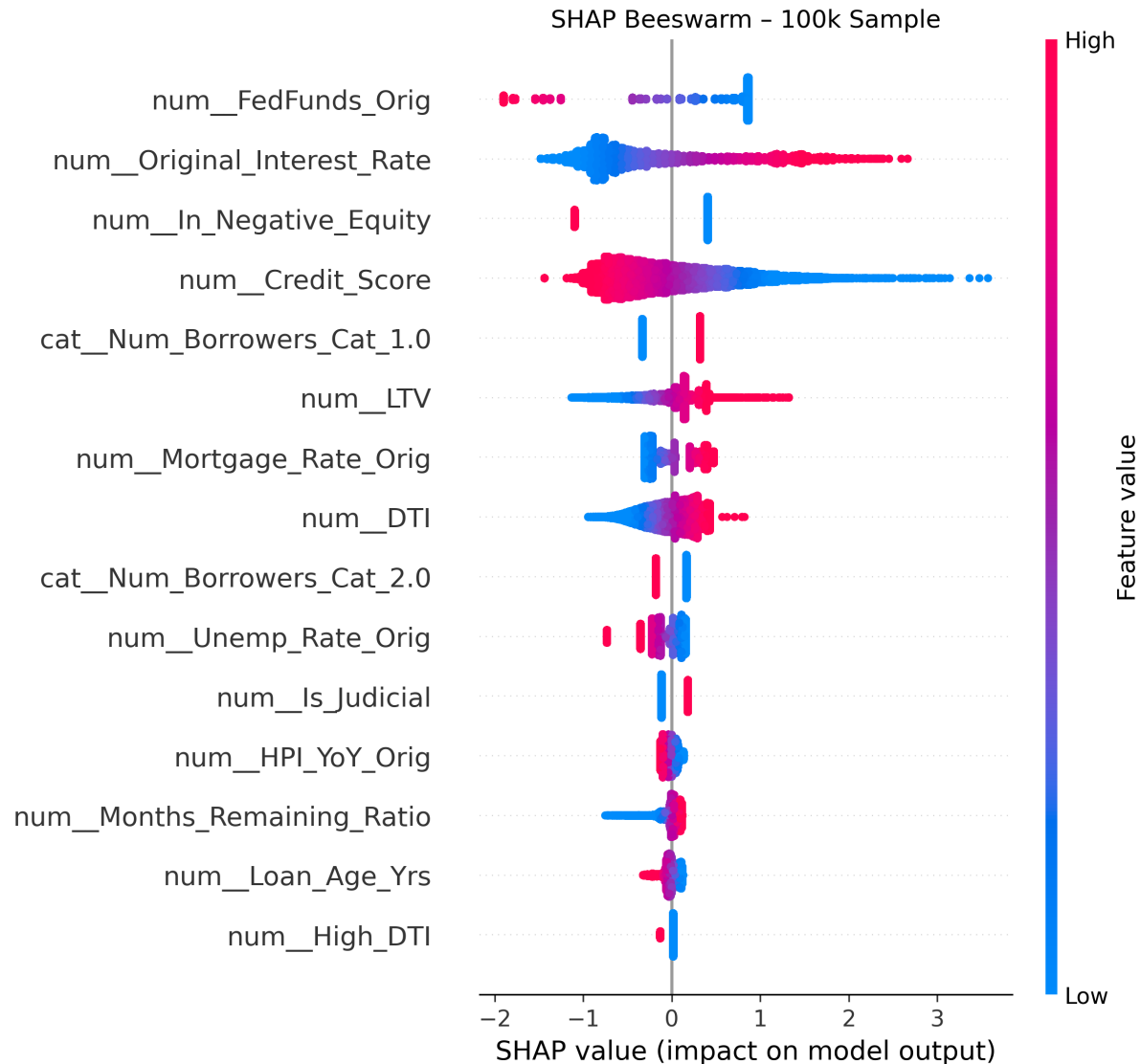
Thank You

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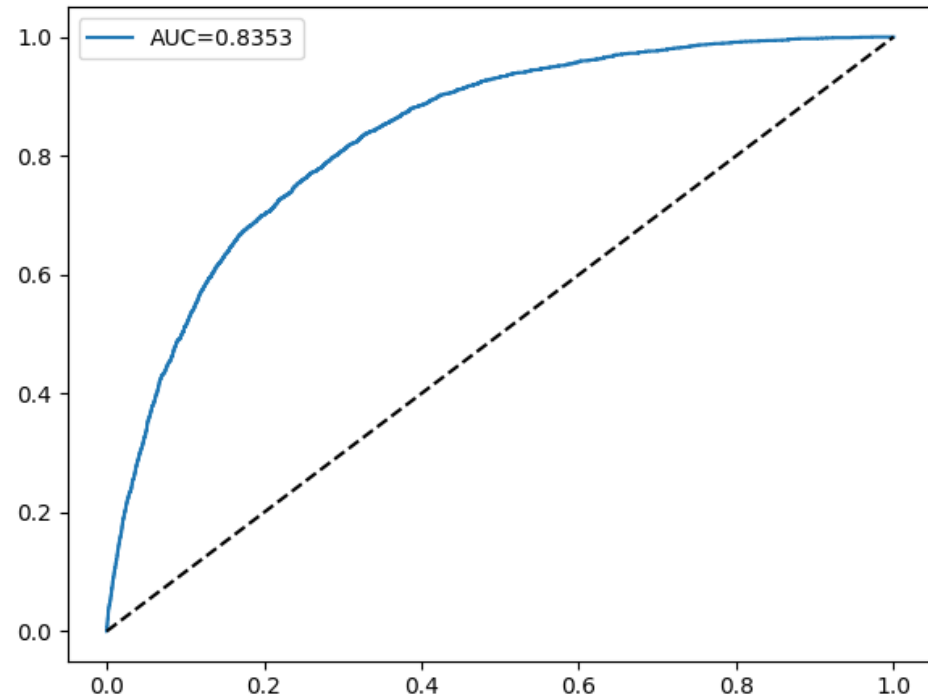
Questions?

SHAP Analysis



AUC Curves

ROC Curve



PR Curve

